

2520030408 Bista Jamuna

Practical 2

Execution:

```
jamuna@Jem:~$ nano practical2.c
jamuna@Jem:~$ gcc practical2.c
jamuna@Jem:~$ ./a.out
File copied successfully
```

```
jamuna@Jem:~$ cat b1
This is the program related to system calls
It is about open read and write system calls
```

```
jamuna@Jem:~$ cat c2
This is the program related to system calls
It is about open read and write system calls
```

```
jamuna@Jem:~$ nano practical2.c
jamuna@Jem:~$ gcc practical2.c
jamuna@Jem:~$ ./a.out
File copied successfully
jamuna@Jem:~$ cat c2
This is the program related to system calls
It is about open read and write system calls
```

```
jamuna@Jem:~$ echo "Hello World" > sample.txt
jamuna@Jem:~$ cat sample.txt
Hello World
jamuna@Jem:~$ strace cat sample.txt
```

```
mmunaJem:"$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffc69eb63e8 /* 25 vars */) = 0
brk(NULL)                                = 0x5de6a1b81000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x737aad72d000
access("/etc/ld.so.preload", R_OK)       = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20085, ...}) = 0
mmap(NULL, 20085, PROT_READ, MAP_PRIVATE, 3, 0) = 0x737aad728000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x737aad6f5000
mmap(0x737aad6f5000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x737aad6fc000
mmap(0x737aad71d000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x737aad71d000
mmap(0x737aad724000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x737aad724000
mmap(0x737aad726000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x737aad726000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x737aad6c7000
mmap(0x737aad6cb000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x737aad6cb000
mmap(0x737aad6ef000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x737aad6ef000
mmap(0x737aad6f3000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x737aad6f3000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1209152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x737aad5a1000
mmap(0x737aad5b3000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x737aad5b3000
mmap(0x737aad653000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x737aad653000
mmap(0x737aad6c5000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x737aad6c5000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0 \250\2\0\0\0\0\0....", 832) = 832
pread64(3, "\6\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....", 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....", 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x737aad200000
mmap(0x737aad228000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x737aad228000
mmap(0x737aad3c0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x737aad3c0000
mmap(0x737aad40e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2d0000) = 0x737aad40e000
mmap(0x737aad414000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x737aad414000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x737aad4f5000
mmap(0x737aad4f8000, 507994, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x737aad4f8000
mmap(0x737aad574000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xf000) = 0x737aad574000
mmap(0x737aad59f000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x737aad59f000
close(3)                                 = 0

mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x737aad6f3000
munmap(0x737aad6f3000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x737aad6f3000
arch_prctl(MACH_SET_FS, 0x737aad6f3000) = 0
set_tid_address(&0x737aad6f120)          = 2174
set_robust_list(&0x737aad6f000, 20)     = 0
rt_sigqueue(0x737aad6f000, 8128, 0, 0x3003055) = 0
mprotect(0x737aad6e0000, 16384, PROT_READ) = 0
mprotect(0x737aad6f0000, 8096, PROT_READ) = 0
mprotect(0x737aad6c0000, 8096, PROT_READ) = 0
mprotect(0x737aad6f3000, 8096, PROT_READ) = 0
mprotect(0x737aad720000, 8096, PROT_READ) = 0
mprotect(0x737aad6f21000, 162272, PROT_READ) = 0
mprotect(0x737aad720000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("V9gXaz2v1B8WddXadXbcVxb*", 8, GRND_NONBLOCK) = 0
munmap(0x737aad728000, 20085)            = 0
statsfs("/sys/fs/selinux", {ftype=VSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffrees=0, f_fsid={val=[0x252f4de9, 0x43323bc]}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_...}, 0) = 0
fcntl(C, F_GETFD)                        = 0
fcntl(C, F_GETFL)                        = 0
fcntl(C, F_GETFD)                        = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0644, st_size=0, ...}) = 0
read(3, "nodeVsSysfs|nnodeVttmpfs|nnodeVtbd...", 1024) = 442
leech(0, SEEK_CUR)                       = 442
leech(0, SEEK_SET)                       = 416
close(3)                                 = 0
openat(AT_FDCWD, "/proc/self/status", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.8.33.2-... 1024) = 1024
read(3, " * 0 @udevpts /dev/pts devpts rw,... 1024) = 1024
read(3, "squid,nodev,noexec,rw,latime,nosy..., 1024) = 1024
read(3, "/usr/1000 tmpfs rw,nosuid,node..., 1024) = 100
read(3, "", 1024)                         = 0
close(3)                                 = 0
access("/etc/selinux/config", F_OK)      = -1 ENOENT (No such file or directory)
fcntl(C, F_GETPD)                        = 0
fcntl(C, F_GETFD)                        = 0
fcntl(C, F_GETPD)                        = 0
rt_sigaction(SIGPROF, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0644, st_size=0, ...}) = 0
read(3, "Sde6da8db5000-Sde6da87800 -p-p ..., 1024) = 1024
read(3, "0 00:30 14176 ..., 1024) = 1024
read(3, "0 00:30 14388 ..., 1024) = 1024
read(3, " /usr/lib/x86_64-lin..., 1024) = 1024
read(3, "0 13931 /us..., 1024) = 568
close(3)                                 = 0
sched_getaffinity(2174, 32, [0 1 2 3 4 5 6 7 8 9 10 11]) = 0
rt_sigaction
```

[illegible]